Requirements

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Requirements Introduction

The initial requirements were identified from the project brief and customer meeting then further worked on by research into similarly styled games and group discussions to ensure that our project meets its intended purpose.

The requirements are categorised as functional requirements, which explains the services and behaviours the game must provide; quality requirements, which define the software qualities and the expected user experience; constraints, which points out restrictions on the technology, design and delivery of the product. Each requirement will be assigned a unique identifier to support traceability throughout design, implementation and testing.

Most of the requirements are aimed at making sure that our project meets the main goal of being user-friendly and family-friendly, suitable for promoting the university of York. They may be refined as the project progresses and further decisions are made by the team.

Functional Requirements

Input/Output Requirements

FR-01 The game will feature simple “WASD” movement which is widely used among other video games to allow for seamless accessibility.

FR-02 The game shall display a top-down view of a university-themed campus

FR-03 The player shall be restricted from moving beyond the map boundaries or through buildings

FR-04 The Geese shall move towards the player and crowd them.

FR-05 The Geese shall enter the playable area from designated spawn points

FR-06 The game shall include a stress bar that drops when the geese touches the player.

FR-07 The game shall include a countdown timer starting at 5 minutes when gameplay starts.

FR-08 The player shall be able to pause and resume the game using the designated keyboard controls

FR-09 The game shall have two starting inventory items with distinct gameplay effects

FR-10 The player shall be able to activate inventory items using designated number keys

FR-11 The game shall display counters for items used and hidden events triggered.

FR-12 The game shall display a victory screen when the agreed victory conditions are met.

FR-13 The game shall provide an agreed outcome when the timer reaches zero or the player fails

FR-14 Restart option/return to menu/character switch/settings, options to be made available while the game is paused and also when the game ends.

FR-15 The game shall provide visual or audio feedback when an item is collected where applicable.

FR-16 The game shall contain hidden events that remain invisible until the player triggers them.

FR-17 The game shall start in a paused state.

Quality Requirements

QR-01 Usability: The game shall provide instructions that allow a new player to understand the controls and begin playing.

QR-02 Maintainability: The code used for the game shall use meaningful class, method and variable names, consistent formatting and the responsibilities clearly separated. The software shall be so that avatars, maps, geese, scoring and other game elements can be changed or extended without extensive changes to unrelated components.

QR-03 Testability: Each system requirement should have at least one associated test case with an expected result.

QR-04 Reliability: The game built must be able to complete a full five minute playthrough without crashing or freezing.

QR-05 Correctness: The hidden event counters, timer, inventory counters will display values that match the actions performed.

QR-06 Usability: A first-time player should be able to start, control, pause and use the game using the instructions provided.

QR-07 Accessibility: The game should contain readable text and should not rely only on colour to communicate important information. Things that have to do with visual interaction should have suitable alternative cues where needed.

QR-08 Inclusivity: The game should be easy to use by players with different levels of gaming experience. It should provide a family-friendly experience for a wide range of audience.

QR-09 Consistency: The visual representation, environment, characters and gameplay effects should remain consistent throughout the game.

QR-10 Legal Compliance: All third-party assets, music, libraries and other resources used should have licences permitting their intended use.

QR-11 Deployment Compatibility: The game shall run on the intended computer environment and view appropriately in full-screen mode.

Constraint Requirements

CR-01 The game shall be coded and developed using Java 17.

CR-02 The game shall support one player per session.

CR-03 The game shall be built using a university inspired setting representing the University of York.
CR-04 The first version of the game shall use the same map layout across the playthroughs.

CR-05 The game shall be non-violent and shall not depict geese or other animals being harmed.

CR-06 The game shall not store player progress, scores or gameplay data between sessions.

CR-07 All third party assets, audio and software libraries shall have licences allowing their intended use and distribution.

CR-08 The game shall support keyboard input and mouse input where required by the implemented controls.

CR-09 The game shall be designed for computer use and support full screen display on a standard monitor.

CR-10 The game shall be delivered as one executable JAR(Java ARchive) that includes all dependencies.